# Bandbreite der Programmiersprachen

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# 99 Bottles of Beer

*99 bottles of beer on the wall, 99 bottles of beer. Take one down and pass it around, 98 bottles of beer on the wall.*
*98 bottles of beer on the wall, 98 bottles of beer. Take one down and pass it around, 97 bottles of beer on the wall.*
*[. . . ]*
*2 bottles of beer on the wall, 2 bottles of beer. Take one down and pass it around, 1 bottle of beer on the wall.*
*1 bottle of beer on the wall, 1 bottle of beer. Take one down and pass it around, no more bottles of beer on the wall.*
*No more bottles of beer on the wall, no more bottles of beer. Go to the store and buy some more, 99 bottles of beer on the wall.*

Quelle: nach "Lyrics of the song 99 Bottles of Beer" on 99-bottles-of-beer.net

# Imperativ, Hardwarenah: C

```c
#define MAXBEER (99)
void chug(int beers);
main() {
   register beers;
   for(beers = MAXBEER; beers; chug(beers--))  puts("");
   puts("\nTime to buy more beer!\n");
}
void chug(register beers) {
   char howmany[8], *s;
   s = beers != 1 ? "s" : "";
   printf("%d bottle%s of beer on the wall,\n", beers, s);
   printf("%d bottle%s of beeeeer . . . ,\n", beers, s);
   printf("Take one down, pass it around,\n");
   if(--beers) sprintf(howmany, "%d", beers); else strcpy(howmany, "No more");
   s = beers != 1 ? "s" : "";
   printf("%s bottle%s of beer on the wall.\n", howmany, s);
}
```

Quelle: "Language C" by Bill Wein on 99-bottles-of-beer.net

# Imperativ, Objektorientiert: Java

```java
class bottles {
    public static void main(String args[]) {
        String s = "s";
        for (int beers=99; beers>-1;) {
            System.out.print(beers + " bottle" + s + " of beer on the wall, ");
            System.out.println(beers + " bottle" + s + " of beer, ");
            if (beers==0) {
                System.out.print("Go to the store, buy some more, ");
                System.out.println("99 bottles of beer on the wall.\n");
                System.exit(0);
            } else
                System.out.print("Take one down, pass it around, ");
            s = (--beers == 1)?"":"s";
            System.out.println(beers + " bottle" + s + " of beer on the wall.\n");
        }
    }
}
```

Quelle: “Language Java” by Sean Russell on 99-bottles-of-beer.net

# Logisch: Prolog

```
bottles :-
    bottles(99).

bottles(1) :-
    write('1 bottle of beer on the wall, 1 bottle of beer,'), nl,
    write('Take one down, and pass it around,'), nl,
    write('Now they are all gone.'), nl,!.
bottles(X) :-
    write(X), write(' bottles of beer on the wall,'), nl,
    write(X), write(' bottles of beer,'), nl,
    write('Take one down and pass it around,'), nl,
    NX is X - 1,
    write(NX), write(' bottles of beer on the wall.'), nl, nl,
    bottles(NX).
```

Quelle: "Language Prolog" by M@ on 99-bottles-of-beer.net

# Funktional: Haskell

```haskell
bottles 0 = "no more bottles"
bottles 1 = "1 bottle"
bottles n = show n ++ " bottles"

verse 0   = "No more bottles of beer on the wall, no more bottles of beer.\n"
         ++ "Go to the store and buy some more, 99 bottles of beer on the wall."

verse n   = bottles n ++ " of beer on the wall, " ++ bottles n ++ " of beer.\n"
         ++ "Take one down and pass it around, " ++ bottles (n-1)
                                                  ++ " of beer on the wall.\n"

main      = mapM (putStrLn . verse) [99,98..0]
```

Quelle: “Language Haskell” by Iavor on 99-bottles-of-beer.net

# Brainfuck

Quelle: Screenshot of “Language Brainfuck” by Michal Wojciech Tarnowski on 99-bottles-of-beer.net

# Programmiersprache Lox

```
fun fib(x) {
    if (x == 0) {
        return 0;
    } else {
        if (x == 1) {
            return 1;
        } else {
            fib(x - 1) + fib(x - 2);
        }
    }
}

var wuppie = fib;
wuppie(4);
```

# Wrap-Up

- Compiler übersetzen formalen Text in ein anderes Format
- Berücksichtigung von unterschiedlichen
  - Sprachkonzepten (Programmierparadigmen)
  - Typ-Systemen
  - Speicherverwaltungsstrategien
  - Abarbeitungsstrategien

# LICENSE

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**Exceptions**

Citations of “99 Bottles of Beer” lyrics and programming language examples (license(s) unknown):

- “Lyrics of the song 99 Bottles of Beer” on 99-bottles-of-beer.net (Author unknown, Website hosted by Oliver Schade, Gregor Scheithauer, and Stefan Scheler)
- “Language C” by Bill Wein on 99-bottles-of-beer.net
- “Language Java” by Sean Russell on 99-bottles-of-beer.net
- “Language Prolog” by M@ on 99-bottles-of-beer.net
- “Language Haskell” by Iavor on 99-bottles-of-beer.net
- Screenshot of “Language Brainfuck” by Michal Wojciech Tarnowski on 99-bottles-of-beer.net (file `screenshot_brainfuck_99bottles.png`)